



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

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Department of Computer Science and Engineering DevOps Lab - L3

Lab Assignment No: 15	
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Class	CSE – B1
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Title of Assignment	Basic Infrastructure as Code (IaC) with Terraform
Implementation & Theory	<pre>PS E:\DevOps_SEM_7> terraform --version Terraform v1.15.8 on windows_amd64 Your version of Terraform is out of date! The latest version is 1.16.1. You can update by downloading from https://developer.hashicorp.com/terraform/install PS E:\DevOps_SEM_7></pre> Terraform Installation: Terraform was successfully installed and verified using the terraform --version command. Terraform is an Infrastructure as Code (IaC) tool used to define and manage infrastructure through declarative configuration files. <pre>Welcome X main.tf U X Assignment - 15 > my-terraform-iac > main.tf 1 terraform { 2 required_providers { 3 local = { 4 source = "hashicorp/local" 5 version = "2.4.0" 6 } 7 } 8 } 9 10 resource "local_file" "my_example_file" { 11 filename = "\${path.module}/example.txt" 12 content = "Hello, Terraform IaC!" 13 }</pre> Terraform Configuration: The main.tf file was created to define the Infrastructure as Code configuration. It uses the hashicorp/local

provider and defines a local_file resource named my_example_file, which creates an example.txt file containing the text "Hello, Terraform IaC!".

```
PS E:\DevOps_SEM_7\Assignment - 15\my-terraform-iac> terraform init
```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

Terraform Initialization: The terraform init command initialized the Terraform working directory and downloaded the required hashicorp/local provider. This prepares the project directory for planning and applying the infrastructure configuration.

```
PS E:\DevOps_SEM_7\Assignment - 15\my-terraform-iac> terraform plan
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# local_file.my_example_file will be created
+ resource "local_file" "my_example_file" {
  + content          = "Hello, Terraform IaC!"
  + content_base64sha256 = (known after apply)
  + content_base64sha512 = (known after apply)
  + content_md5       = (known after apply)
  + content_sha1       = (known after apply)
  + content_sha256     = (known after apply)
  + content_sha512     = (known after apply)
  + directory_permission = "0777"
  + file_permission    = "0777"
  + filename           = "./example.txt"
  + id                 = (known after apply)
}
```

Plan: 1 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

```
PS E:\DevOps_SEM_7\Assignment - 15\my-terraform-iac>
```

Terraform Execution Plan: The terraform plan command was executed to preview the infrastructure changes before applying them. Terraform detected that one new local file resource would be created, resulting in a plan of 1 resource to add, 0 to change, and 0 to destroy.

```
PS E:\DevOps_SEM_7\Assignment - 15\my-terraform-iac> terraform apply
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# local_file.my_example_file will be created
+ resource "local_file" "my_example_file" {
  + content          = "Hello, Terraform IaC!"
  + content_base64sha256 = (known after apply)
  + content_base64sha512 = (known after apply)
  + content_md5       = (known after apply)
  + content_sha1       = (known after apply)
  + content_sha256     = (known after apply)
  + content_sha512     = (known after apply)
  + directory_permission = "0777"
  + file_permission    = "0777"
  + filename           = "./example.txt"
  + id                 = (known after apply)
}
```

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

local_file.my_example_file: Creating...

local_file.my_example_file: Creation complete after 0s [id=67c458d520445706ef3f2d7483fd5f652f7ba55b]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

```
PS E:\DevOps_SEM_7\Assignment - 15\my-terraform-iac> Get-Content example.txt
```

```
Hello, Terraform IaC!
```

```
PS E:\DevOps_SEM_7\Assignment - 15\my-terraform-iac>
```

Terraform Apply: The terraform apply command was used to create the infrastructure defined in main.tf. After confirmation, Terraform created example.txt. The file content was verified successfully, confirming that the local file resource was created as defined in the Terraform configuration.

```
main.tf U X
Assignment - 15 > my-terraform-iac > main.tf
1 terraform {
2   required_providers {
3     local = {
4       source = "hashicorp/local"
5       version = "2.4.0"
6     }
7   }
8 }
9
10 resource "local_file" "my_example_file" {
11   filename = "${path.module}/example.txt"
12   content = "Hello, Terraform IaC! Updated successfully."
13 }
```

Terraform Configuration Modified: The content of the local_file resource was modified in main.tf. This change demonstrates how Infrastructure as Code allows infrastructure resources to be updated simply by modifying the declarative configuration.

```
PS E:\DevOps_SEM_7\Assignment - 15\my-terraform-iac> terraform plan
local_file.my_example_file: Refreshing state... [id=67c458d620445706ef3f2d7483fd5f652f7ba55b]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
~/. destroy and then create replacement

Terraform will perform the following actions:

# local_file.my_example_file must be replaced
~/. resource "local_file" "my_example_file" {
  ~ content      = "Hello, Terraform IaC!" -> "Hello, Terraform IaC! Updated successfully." # forces replacement
  ~ content_base64sha256 = "Hl3JlUWwE7TPM9KCELt43/Dn6E7rjUJtchBm3Juh" -> (known after apply)
  ~ content_base64sha512 = "Jh1z1QVWwY50e9TvKq3qah1k1zVQF5eS6U889h04juj56b7w0Dclcp9z9T727PKzGmH+Demv02npJtTQ==" -> (known after apply)
  ~ content_md5      = "ea90237e8d9755dbf792df68f4922f" -> (known after apply)
  ~ content_sha1      = "67c458d620445706ef3f2d7483fd5f652f7ba55b" -> (known after apply)
  ~ content_sha256     = "34b5d1948547ade4ca3e14960842ebb78dff8be84ad88ee52d7b4c019bc2543" -> (known after apply)
  ~ content_sha512     = "Fc7ce548868c32fc9239bd2c4ef68aab7a9a1e22e12f35054549e4946c14a1a388eebe3e7a6d3c018de2dca687fdcd34d97cc2b31a637e0de9bee69e926d4d" -> (known after apply)
  ~ id              = "67c458d620445706ef3f2d7483fd5f652f7ba55b" -> (known after apply)
  # (3 unchanged attributes hidden)
}

Plan: 1 to add, 0 to change, 1 to destroy.
```

Modified Execution Plan: After updating the Terraform configuration, terraform plan detected the difference between the existing resource and the desired configuration. Terraform generated a new execution plan showing the required change to update the managed resource.

```

PS E:\DevOps_SEM_7\Assignment - 15\my-terraform-iac> terraform apply
local_file.my_example_file: Refreshing state... [id=67c458d620445706ef3f2d7483fd5f652f7ba55b]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
-/+ destroy and then create replacement

Terraform will perform the following actions:

# local_file.my_example_file must be replaced
-/+ resource "local_file" "my_example_file" {
  ~ content           = "Hello, Terraform IaC!" -> "Hello, Terraform IaC! Updated successfully." # forces replacement
  ~ content_base64sha256 = "NLXR1AVHreTKPhSWCELRt43/Dn6ErYjuUte@w8m83UM=" -> (known after apply)
  ~ content_base64sha512 = "/HzlQIYmVvSOb0stVakQ3qHLLH1VQVfSe5UBBSho4juvj56bTw8DeLcpof9zITZfKzGmVHDem+D2npJITQ=" -> (known after apply)
  ~ content_md5         = "a8b0c23e8d0755dbf72d4fd688f022f" -> (known after apply)
  ~ content_sha1         = "67c458d620445706ef3f2d7483fd5f652f7ba55b" -> (known after apply)
  ~ content_sha256       = "34b5d1940547ade4ca3e14960842abb78dff0be84ad8ee52d7b4c819bc2543" -> (known after apply)
  ~ content_sha512       = "fc7ce540860c32fc9239bd2c4ef68aab7a9a1e22e12f355054549e4946c14a1a388eebe3e7a6d3c010de2dca687fcd34d97cc2b31a637e0de9bae"
  pply) ~ id           = "67c458d620445706ef3f2d7483fd5f652f7ba55b" -> (known after apply)
        # (3 unchanged attributes hidden)
}

Plan: 1 to add, 0 to change, 1 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

Enter a value: yes

local_file.my_example_file: Destroying... [id=67c458d620445706ef3f2d7483fd5f652f7ba55b]
local_file.my_example_file: Destruction complete after 0s
local_file.my_example_file: Creating...
local_file.my_example_file: Creation complete after 0s [id=51a23989728bb7d451b41ac62293d1e5c8645789]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
PS E:\DevOps_SEM_7\Assignment - 15\my-terraform-iac> Get-Content example.txt
Hello, Terraform IaC! Updated successfully.
PS E:\DevOps_SEM_7\Assignment - 15\my-terraform-iac>

```

Terraform Configuration Updated: The modified Terraform configuration was successfully applied. Terraform reconciled the resource with the updated desired state, and the contents of example.txt were verified to confirm the change.

```

PS E:\DevOps_SEM_7\Assignment - 15\my-terraform-iac> terraform destroy
local_file.my_example_file: Refreshing state... [id=51a23989728bb7d451b41ac62293d1e5c8645789]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# local_file.my_example_file will be destroyed
- resource "local_file" "my_example_file" {
  - content           = "Hello, Terraform IaC! Updated successfully." -> null
  - content_base64sha256 = "167/Lt2rHJ0PmPw4d8b1/091V1pyBoduUW463H=" -> null
  - content_base64sha512 = "x2Vx3Q8HdVxKLw@BAtCpD7IOhR814/ARQW4R1Ar@hnsK91VVRP1/shotmELVnKj31rEBVv8FFpAVD3d43DwEq=" -> null
  - content_md5         = "d83e5df7c3c1b3f1fde43a5e9a054a6" -> null
  - content_sha1         = "51a23989728bb7d451b41ac62293d1e5c8645789" -> null
  - content_sha256       = "d7aeff2d36d9aeb31439f60ca56cf86e123ff7d4d5222a7221ba3ab94638e883" -> null
  - content_sha512       = "c7663edce07d676245cb7aaf1a0130a90fb20ef91475e3fe34415e11880ef4867e4af6561544f23fb21b66112260d923de5ac4f15bfc145328543d0da370f01d" -> null
  - directory_permission = "0777" -> null
  - file_permission      = "0777" -> null
  - filename             = ".example.txt" -> null
  - id                   = "51a23989728bb7d451b41ac62293d1e5c8645789" -> null
}

Plan: 0 to add, 0 to change, 1 to destroy.

Do you really want to destroy all resources?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

local_file.my_example_file: Destroying... [id=51a23989728bb7d451b41ac62293d1e5c8645789]
local_file.my_example_file: Destruction complete after 0s

Destroy complete! Resources: 1 destroyed.
PS E:\DevOps_SEM_7\Assignment - 15\my-terraform-iac> Test-Path .\example.txt
False
PS E:\DevOps_SEM_7\Assignment - 15\my-terraform-iac>

```

terraform destroy removes the infrastructure resources managed by Terraform. The False result confirms that the Terraform-created example.txt has been successfully removed.